GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. I Semester 2024-25
1FY2-01: Engineering Mathematics-I
Branch-All
Guess Paper

Part A(Answer should be given up to 25 words only)
All questions are compulsory

Q.1 Find the value of $\Gamma\text{-}\frac{1}{2}$

Q.2 Write Euler's theorem on homogeneous function.

Q.3 Evaluate: $\iint x\ y\ dxdy$, where the region of integration is x+y<1 in the positive quadrant

Q.4 Find the value of $\int_0^\infty e^{-x}x^3dx$

Q.5 Find the value of $\int_0^1 x^3(1-x)^2dx$

Q.6 Find the value of $\int_0^{\pi/2} sin^3x\ cos^5x\ dx$

Q.7 Find the value of $\beta(\frac{1}{2},\frac{3}{2})$.

Q.8 State Stoke's theorem.

Q.9 Find the limit of the sequence $<x_n>$, where $x_n = \frac{5n-3}{7n+8}$

Q.10 Find the first order partial derivative of u = $tan^{-1}\left(\frac{y}{x}\right)$

Q.11 Find a_0 for f(x) = |x|, where $-\pi < x < \pi$.

Q.12 Write the coordinate of Centre of gravity of a solid.

Q.13 Write the condition for p series $\sum_{n=1}^{\infty}\frac{1}{a^n}$ to be convergent and divergent.

Q.14 If $\vec{F} = x^2y\hat{\imath} - 2xy^2\hat{\jmath} + 3x^2z\hat{k}$, find divergence at point (3, -1, -2).

Q.15 State Parseval's theorem.

Q.16 Let f = y^x, what is $\frac{\partial^2 f}{\partial x\partial y}$ at x = 2, y =1?

Q.17 Evaluate $\int_0^b \int_0^x xy\ dx\ dy$

Q.l8 What is the value of integral $\int_0^\infty e^{-x^2}\ dx$?

Q.19 Write the formula of surface areas of solid of revolution when the revolution is about x-axiss

Q.20 What do you mean by convergence of sequence

Q.21 Find whether the following series is convergent or not?

$$\frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \cdots$$

Q.22 Slate Parseval's theorem.

Q.23 Find the value of a_0 for the function f(x) =|x| in the interval $(-\pi, \pi)$.

Q.24 State the necessary and sufficient conditions for the minimum of a function f (x,y),

Q.25 Find the gradient of f(x,y,z) = $x^2y^2 + xy^2 - z^2$at (3,1,1)

Q.26 Evaluate $\int_0^b \int_0^x xydxdy$

Q.27 State the gauss divergence theorem.

Q.28 Write the power series expansion of logarithm function.

Q.29 Evaluate a_n in the Fourier series of the function f(x) = $x + x^2, \pi < x < \pi$.

Q.30 Define Cauchy's (E $-\delta$) definition of continuity.

Q.31. Evaluate: $\int_0^\infty x^6 e^{-2x}\, dx$ by using beta- gamma function

Q.32 Write relation between Beta and Gama functions.

Q.33 Change the order of integration of the following double integration: $\int_0^4 \int_x^{2\sqrt{x}} f(x,y)dxdy$

Q.34 Write the area of a solid of revolution about X-axis in Cartesian form.

Q.35 F=$x^2yi - 2xy^2zj + 3x^2zk$, find divF at the point (3,-1,-2)

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

Q.1 Evaluate $\int_0^\infty \frac{x}{1+x^6} dx$

Q.2 Find the volume of the solid generated by revolving the astroid $x^{2/3} + y^{2/3} = 1$

Q.3 Find the value of $\int_0^{\pi/6} sin^2 6x\ cos^4 3x\ dx$

Q.4 Show that β (m,n) = β (m+1,n) +β(m,n+1).

Q.5 Prove that $\int_0^{\pi/2} tan^n x\, dx = \frac{\pi}{2} sec \frac{n\pi}{2}$

Q.6 Find the Fourier sine series of x in interval 0 < x < π.

Q.7 The directional derivative of scalar function f(x, y, z) = $x^2 + 2y^2 + z$ at the point P(1, 1, 2) in the direction of the vector $\vec{a} = 3\hat{\imath} - 4\hat{\jmath}$.

Q.8 Evaluate: $\iiint_R (x+y+z)dxdydz$ where R: $0 \le x \le 1$, $1 \le y \le 2$ and $2 \le z \le 3$.

Q.9 Test for the convergence for series: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

Q.10 If f(x, y) = $20x + 26y + 4xy - 4x^2 - 3y^2$, find (x, y) to maximize.

Q.11 Expand sin x in the power of (x $-\frac{\pi}{2}$).

Q.12 Find the Fourier series of x^2 in $(-\pi, \pi)$, and use parseval identity to prove $\frac{\pi^4}{90} = 1 + \frac{1}{2^4} + \frac{1}{3^4} + \ldots$

Q.13 If u= e^{xyz} , then show that: $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$

Q.14 Whether the fluid motion by given by V=$(y+z)i + (z+x)j + (x+y)k$ *is incompressible or not* ?

Q.15 Change the order of integration and hence evaluate: $\int_0^1 \int_{e^x}^{e} \frac{1}{logy} dxdy$

Q.16 Evaluate $\int_1^2 \int_1^z \int_0^{yz} (xyz)dxdydz$

Q.17 Test the convergence of the following series $-\frac{1}{2} + \frac{1.3}{2.4} + \frac{1.3.5}{2.4.6}$ ... .

Q.18 Find a Fourier series for the function f(x) = x^2 in the interval $-\pi < x < \pi$ and deduce the following term

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi}{12}$$

Q.19 Find the equations of the tangent plane and normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point (1, 2,-1).

Q.20 Evaluate the point where the function- $\boldsymbol{x^3}y^2(1-x-y)$ Will have maxima. Also find the maximum value.

Q.21 Evaluate the integral - $\int_0^1 \int_0^x \frac{x^3 dxdy}{\sqrt{x^2+y^2}}$ by changing into polar coordinates

Q.22 If a and bare differentiable vector point functions. then prove that div (a+b) = b curl a- a.curl b

Part C(Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions(word Limit 125)

Q.1 Find the surface area of the solid formed by revolving the ellipse $\frac{x^2}{a^2}+\frac{x^2}{b^2}=1$about the major axis.

Q.2. Prove that $\Gamma\frac{1}{2}=\sqrt{\pi}$

Q.3 Find the surface area of the solid formed by revolving the cardioid r = a(1+cosθ) about the initial line.

Q.4 Test for the convergence for series: $\frac{1.2}{3^2 4^2}+\frac{3.4}{5^2 6^2}+\frac{5.6}{7^2 8^2}+\cdots$

Q.5. Find half range cosine series for the function f(x) = 2x − 1.

Q.6 Apply Stoke's theorem to find the value of $\int_C \mathrm{ydx}+\mathrm{zdy}+\mathrm{xdz}$ where c is the intersection of $x^2 + y^2 + z^2 = a^2$ and x + z = a.

Q.7 Use beta and gamma function to evaluate: 1. $\int_0^\infty \frac{x}{1+x^6}$ 2. $\int_0^1 \sqrt{\frac{1-x}{x}}\,dx$

Q.8 Use Lagrange's method to find the maximum and minimum distance of the point (3, 4, 12) from the sphere $x^2+y^2+z^2=1$.

Q.9 If u=f(r) , where $r^2=x^2+y^2$:then prove that:

$\frac{\delta^2 u}{\delta x^2}+\frac{\delta^2 u}{\delta y^2}=f''(r)+\frac{1}{r}f'(r)$

Q.10 Verify greens theorem for $\int_c[(xy+y^2)dx+x^2dy]$, where C is the closed curve of the region bounded by y = x and y = x^2

Q.11 Find the volume of spindle shaped solid generated by revolving the Astroid about the x-axis – $x^{2/3} + y^{2/3} = a^{2/3}$

Q.12 Find half range cosine series for the function: f(x) = 2x – 1 ,0 < x < 1 hence deduce that $\frac{\pi^2}{8}=\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}$....

Q.13 Find the volume of the tetrahedron bounded by the co - ordinate planes and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

Q.14 State Gauss's divergence theorem. Verify Gauss's divergence theorem for F = xyi +z^2 j + 2yz k on the tetrahedron x = y = z = 0 and x + y + z = l.